7. Part-of-Speech Tagging Using NLTK

Aim: To perform Part-of-Speech (POS) tagging using the NLTK library.

Procedure:

1. Install NLTK.
2. Import the NLTK library.
3. Download the required language data.
4. Tokenize the sentence into words.
5. Assign and display POS tags.

Code:

```
import nltk

nltk.download('averaged_perceptron_tagger_eng')

nltk.download('punkt_tab')

text = "The boy plays cricket"

words = nltk.word_tokenize(text)

tags = nltk.pos_tag(words)

print(tags)
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/7.py"
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     C:\Users\blohi\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\blohi\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[('The', 'DT'), ('boy', 'NN'), ('plays', 'VBZ'), ('cricket', 'NN')]
```

Result: POS tagging was successfully performed using NLTK.